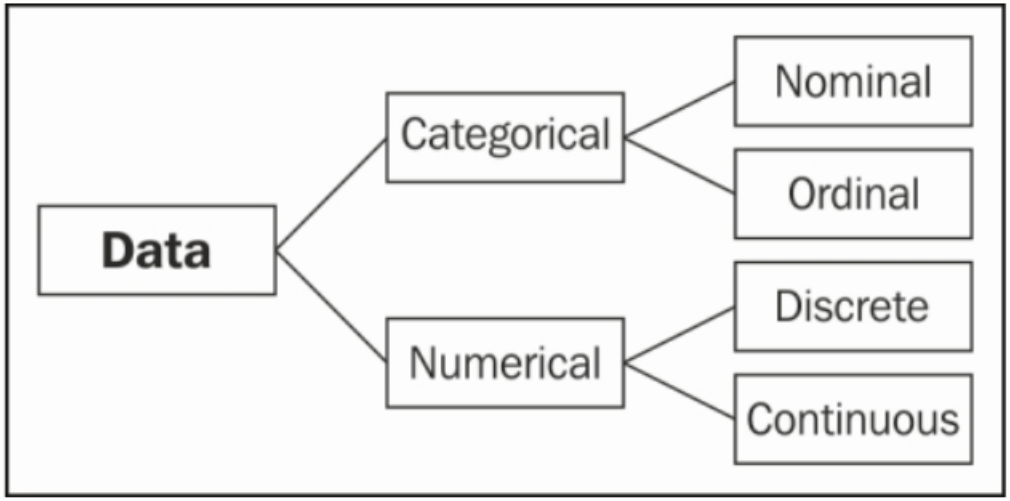
The nature of data

Data is the plural of datum, so it is always treated as plural. We can find data in all the situations of the world around us, in all the structured or unstructured, in continuous or discrete conditions, in weather records, stock market logs, in photo albums, music playlists, or in our Twitter accounts. In fact, data can be seen as the essential raw material of any kind of human activity. According to the Oxford English Dictionary: ***Data are known facts or things used as basis for inference or reckoning.***

As shown in the following figure, we can see **Data** in two distinct ways: **Categorical** and **Numerical**:



1. Categorical Data:

Categorical data are values or observations that can be sorted into groups or categories. There are two types of categorical values, nominal and ordinal.

1.1. A nominal variable has no intrinsic ordering to its categories. For example, housing is a categorical variable having two categories (own and rent).

1.2. An ordinal variable has an established ordering. For example, age as a variable with three orderly categories (young, adult, and elder).

2. Numerical data:

Numerical are values or observations that can be measured.

There are two kinds of numerical values, discrete and continuous.

2.1. Discrete data are values or observations that can be counted and are distinct & separate. Discrete data involves round, concrete numbers that are determined by counting.

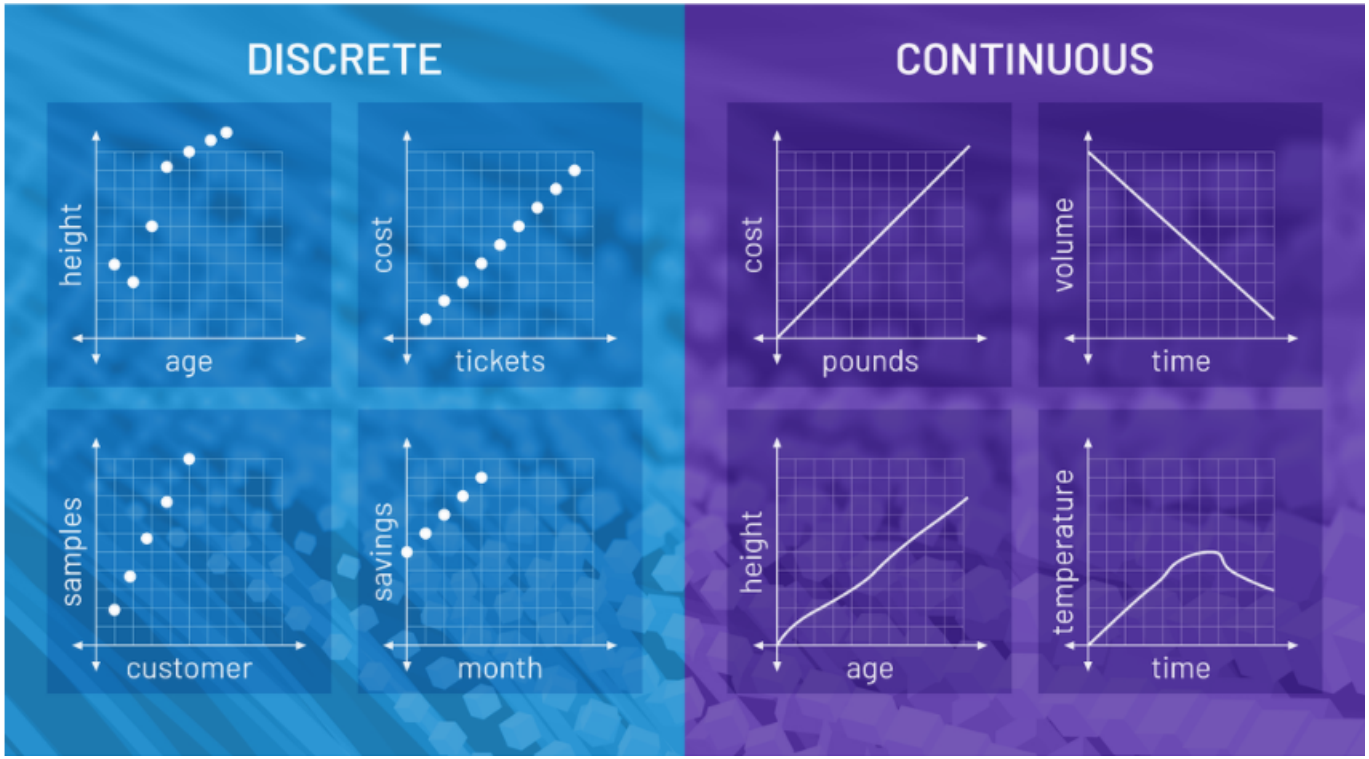
The discrete data are finite or potentially countable set of values. For example, the number of members in a library. It can be 2,575 or 2,599 but certainly not 2,599½.

Similarly, the number of citizens in a country, the number of vehicles registered is the examples of discrete data.

Some other examples of discrete data

- The number of customers who bought different items
- Number of lines in a code.
- The number of computers in each department
- The number of items you buy at the grocery store each week

Discrete Vs Continuous Data



2.2. Continuous data are values or observations that may take on any value within a finite or infinite interval. For example, an economic time series such as historic gold prices, Height, weight, temperature and length.

Continuous data are an infinite set of possible values. Between a range there are infinite possible values.

For example, height of an individual is not restricted to values like 155 cm. and after that to 156 cm. It can be 155.59 cm. or 155.99 cm. – continuous value.

Continuous data refers to the unfixed number of possible measurements between two realistic points.

Continuous data is all about accuracy.

Variables in these data sets often carry decimal points, with the number to the right stretched out as far as possible. This level of detail is paramount for scientists, doctors and manufacturers, just to name a few.

Continuous variables can take on almost any numeric value and can be meaningfully divided into smaller increments, including fractional and decimal values.

You often measure a continuous variable on a scale. For example, when you measure height, weight, and temperature, you have continuous data.

Examples of continuous data

- The weight of newborn babies
- The daily wind speed
- The temperature of a freezer